



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0915 PROGRAMMING IN JAVA

Java-Based Computer Recommendation & Performance Analysis System

Presented by:

K. Vijay Bhaskar Reddy (192425155)

Guided by:

MEGANATHAN E

Project Overview

Bridging Workloads with Optimal Hardware

Selecting an optimal computer configuration requires matching complex user applications with rapidly changing hardware architectures.

- 1. Workload Ingestion:** Captures user budget, CPU usage, RAM, and storage demands.
- 2. Hardware Mapping:** Expert decision engine recommends CPU, GPU, and memory setups.
- 3. Performance Analytics:** Calculates execution time, power usage, and PDF summaries.

Problem Statement



Static Pre-builds

Generic PC recommendation engines rely on rigid pre-configured lists that fail to adapt to dynamic workload demands.



Resource Mismatch

Users misallocate budget by overspending on unnecessary specs while underpowering critical application bottlenecks.



No Analytics

Traditional tools recommend hardware without offering execution time estimation, TDP power calculations, or reports.

Key Objectives

- 🎯 **Requirement Profiling:** Collect budget, workload class, CPU, RAM, and storage parameters.
- 🎯 **Rule-Based Recommendation:** Build an expert matrix to recommend optimal CPU cores, cache, RAM, and GPU.
- 🎯 **Performance Evaluation:** Calculate execution time estimates, TDP wattage, and overall architecture score.
- 🎯 **Automated Reporting:** Export interactive charts and downloadable PDF diagnostic summaries.

⚙️ Rule Engine Matrix

100%

Automated mapping of complex user parameters into concrete, non-bottlenecked system hardware configurations.

System Architecture Framework

MODULE 1

Requirement Profiling

Collects budget, workload type, CPU/RAM/Storage intensity, and creates structured requirement profiles.

Tech: Python, Streamlit, SQLite

MODULE 2

Architecture Design

Rule-based decision matrix mapping CPU cores, clock speed, cache size, RAM capacity, and GPU requirements.

Tech: Rule-Based Expert System

MODULE 3

Performance & Reports

Evaluates scores, estimates execution time and power usage, offering dashboard charts & PDF reports.

Tech: Matplotlib, ReportLab

Summary Table of System Modules

Module Name	Core Objective	Key Inputs / Specs	Output & Tech Stack
Module 1: Requirement Profiling	Collect user parameters & analyze workload needs	Budget, workload type, CPU, RAM, storage, app type	Structured requirement profile
Module 2: Architecture Builder	Generate optimal computer architecture configuration	CPU cores, clock speed, cache, RAM, storage, GPU	Optimized system configuration
Module 3: Performance Module	Evaluate configuration and generate analytics reports	Score, execution time, power usage (TDP), upgrade score	Interactive charts & PDF report

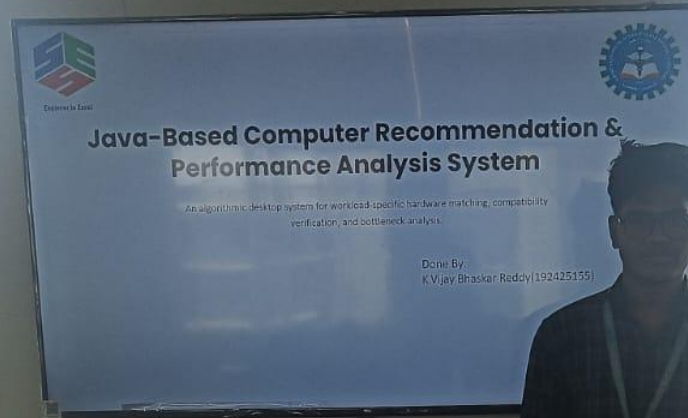
Key Advantages & Future Scope

★ System Advantages

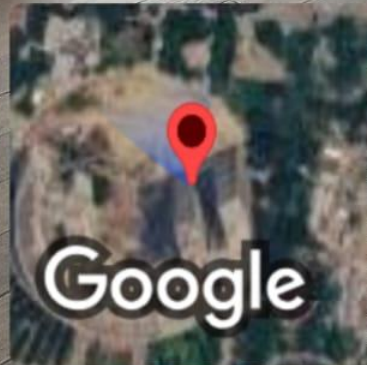
- ✓ **Workload Efficiency:** Eliminates hardware mismatch and bottlenecks.
- ✓ **Cost Optimization:** Maximizes performance within budget limits.
- ✓ **Comprehensive Reports:** Downloadable PDF performance summaries.

✚ Future Scope

- > **Live Pricing APIs:** Fetch real-time hardware component prices.
- > **Machine Learning:** Predictive models for workload profiling.
- > **Hardware Validation:** Automated socket & motherboard VRM checks.



 GPS Map Camera



Mevalurkuppam, Tamil Nadu, India



22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India

Lat 13.02662° Long 80.016946°

Friday, 31/07/2026 11:56 AM GMT +05:30